OBSERVABILITY AND TRACING EVIDENCE

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening | Week 7 | Aryampa Joshua

1. Purpose

This note is the hand-in for the Week 7 Observability and tracing evidence deliverable. Each v1.3 student turn writes one JSON record that covers model, prompt, retrieval, tools, latency or error, and outcome.

2. What was added

- src/observability/turn_log.py - builds and writes the turn record.
- src/tools/orchestrate.py - calls the logger on every turn exit, including Sense stops and model failures.
- logs/ - runtime turn files (gitignored).
- src/traces/run_week7_observability.py - regenerates the checked-in samples.
- evidence/traces/O1-O4 - the four samples below.

3. Fields in each turn log

- Identity: turn_id, session_id, started_at, ended_at.
- Model: name (gemini-3.6-flash) and rounds in that turn.
- Prompt: id (v1.3) and whether a Case History Recall note was added. Full prompt text is not stored.
- Retrieval: query, status, count, and source titles with similarity scores. Passage bodies are not stored.
- Tools: name, arguments, and a shortened result for every tool the app ran.
- Latency and errors: latency_ms, and error when the model call fails.
- Outcome: stop_condition, draft_saved, active_case_id, and a short reply_preview.

4. Evidence samples

O1_blocked_topic.json
Admissions question. Sense stops at S5 before the model is called. Tools and retrieval are empty.

O2_adversarial_refusal.json
Instruction-override attempt. Sense stops at S6. No tools run.

O3_grounded_answer.json
Real retrieve_documents result (3 sources) written through the Week 7 logger. Gemini returned 503 during the full turn, so this sample is retrieval-backed. Re-run O3 when the API recovers for a full model path.

O4_model_unavailable.json
Live generate_content call failed with 503 UNAVAILABLE. latency_ms and error are set. No tools ran.

5. How to re-run

From the repository root: python src/traces/run_week7_observability.py

Optional arguments: O1 O2 O3 O4. Live model turns need GEMINI_API_KEY in .env. If Gemini

returns 503, O3 falls back to logging a real retrieve_documents result; re-run O3 when the API recovers for a full model path.

6. Limits of this evidence

Routine turn files stay under logs/ and are not committed. The four O-samples are the durable evidence. Full system prompts and retrieved passage text are omitted on purpose so the files stay small and do not copy corpus text into Git.